

Arushi Dodhia

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Detail-oriented and versatile Software Engineer with hands-on experience across full-stack development, machine learning integration, and mobile application design. Proven ability to architect scalable backend systems using Python (Flask, FastAPI), build responsive frontends with React.js, and develop Android applications using Kotlin. Skilled in database design (SQL, NoSQL), RESTful API development, and Agile methodologies. Passionate about creating user-centered digital solutions that merge technical excellence with impactful design, especially in research-driven and mission-oriented environments.

EDUCATION

University of Wisconsin-Madison | Madison, WI

May 2025

Bachelors of Science: Computer Science

- Relevant Coursework: *Programming III, Discrete Mathematics, Linear Algebra with Matrices, Calculus III, Introduction to Computer Engineering, Introduction to Artificial Intelligence, Introduction to Cryptography, Machine Organization and Programming, Introduction to Algorithms, Foundation of Mobile Systems and Applications, Building User Interfaces, Software Engineering*

WORK EXPERIENCE

HCI Research | UW-Madison

May 2025 - Present

Software Engineer Researcher

- Developed scalable RESTful APIs using Flask, FastAPI, and Python, ensuring modular, maintainable backend services.
- Integrated different machine learning models to support dynamic chat features for tasks such as reading comprehension, writing, and math
- Developed APIs using SQL to manage user input and interface logic across multiple interaction modes (e.g., multiple choice, open-ended responses)
- Prototyped database integration for storing and retrieving experiment questions and responses, enabling scalable data management
- Used WebSocket protocols for real-time communication between a Kotlin-based robot client and a web dashboard.
- Deployed features to support experimental user interaction workflows, enabling multi-modal UI (touch, voice, text).
- Created an interactive research project for developmental learning with children, using the robot as an agent for engagement and data collection

The Coder School | Los Gatos, CA

June 2023 – Present

Tutor

- Delivered hands-on instruction in Python, Scratch, and game development, reinforcing foundational computational thinking.
- Led curriculum design and project-based learning around OOP, event-driven programming, and visual programming paradigms.
- Mentored students on debugging techniques, logical reasoning, and building playable games like Flappy Bird and Pac-Man.

Rush Cakes | San Jose, CA

June 2021 – Present

Entrepreneur

- Launched a mission-driven startup with full-stack website implementation using HTML, CSS, and JavaScript.
- Started a non-profit that donates to the National Pediatric Cancer Foundation by selling vegan and eggless cupcakes in the Bay Area
- Executed digital marketing campaigns via SEO and social media platforms, increasing visibility and online sales.
- Managed end-to-end product lifecycle and business operations while integrating e-commerce solutions for order fulfillment.

PROJECTS

Shopbop/Amazon (Capstone)

Jan 2025 - May 2025

- Developed the frontend interface using React.js for an AI-driven outfit matching feature, integrating the Shopbop API to fetch and display clothing recommendations.
- Engineered a swiping functionality using React, allowing users to interact with four individual swipeable cards which maximize on click and a larger dynamically updating card.
- Coded the landing page and swiping page, ensuring enhanced user engagement.
- Collaborated with backend developers to integrate a machine learning algorithm, which analyzes user swipes to refine and personalize outfit recommendations in real time, leveraging behavioral data to improve accuracy.
- Collaborated on cross-functional teams to execute full-stack development, applying Agile/Scrum practices with JIRA and GitHub Projects.

Textbook for Intro to Computer Engineering

September 2024 – Present

- Spearheading the user experience (UX) design for an interactive digital textbook, collaborating with a professor and peers to enhance engagement and concept retention.
- Developing dynamic, interactive simulations using React and JavaScript to provide hands-on learning experiences, reinforcing fundamental computer engineering principles for topics like binary conversion, addition, ICEE 754 conversion, and logic gates.
- Engineering an intuitive and visually compelling interface to optimize usability and streamline content accessibility for diverse learners.

SecondServe

September 2024 – December 2024

- Developed a cross-platform app with Kotlin (frontend) and Python Flask backend, using MVVM architecture and Room DB.
- Ensured smooth navigation and state management with Jetpack Compose and LiveData.
- Integrated Google Maps API and Firebase Authentication, enabling geolocation and secure user sessions.

School Project - LeNet

November 2023

- Engineered a customized Convolutional Neural Network (CNN) using PyTorch and deep learning frameworks to improve image recognition accuracy.
- Optimized model performance through hyperparameter tuning, refining batch size, epochs, and learning rate to maximize efficiency and precision.
- Implemented advanced machine learning techniques to enhance feature extraction, achieving improved classification accuracy and model generalization.

CERTIFICATES

Biology Meets Programming: Bioinformatics for Beginners | University of California, San Diego

January 2025

- Gained skills in bioinformatics algorithms and Python programming for DNA sequence analysis and motif finding.

Databases and SQL for Data Science with Python | IBM

July 2024

- Proficient in analyzing databases using SQL and Python, creating relational databases, working with multiple tables using DDL commands, constructing basic to intermediate level SQL queries using DML commands, and composing more powerful queries with advanced SQL techniques like views, transactions, stored procedures, and joins.

Crash Course on Python | Google

October 2020

- Proficient in Python for automation and script writing; skilled in using strings, lists, dictionaries, and creating custom Python objects.

SKILLS & INTERESTS

Programming languages, frameworks, and tools: Java, Python, JavaScript, React.js, Node.js, Flask, Kotlin, SQL, HTML, CSS, REST API, Docker, Agile, Git, React Native, TypeScript, C, R, Swift, Pandas, Flask, FastAPI, PyTorch, MongoDB, PostgreSQL, GitHub, GitLab, Bash, Gradle

Software development concepts: Object-Oriented Programming (OOP), Data Structures, Algorithms, Testing, CI/CD, Version Control, API Design, AI/ML

Tools & technologies: GitHub, Postman, WebSockets, SQL databases, NoSQL, Agile/Scrum

Languages: English (Fluent), Gujarati (Advanced), Hindi (Advanced), French (Intermediate), Spanish (Intermediate)

Clubs: UW Cybersecurity, Leet Code Club, Women in STEM, Wisconsin Range (Recreational Bollywood Dance Team)